

Beat the Sugar Spike

Real-Time, Context-Aware Nudges for Healthier Choices

Problem Statement

Prepared for Hackathon Submission

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Chapter 1

The Problem & The Solution

1.1 Problem Description

Many young people (ages 16–32) want to live healthier lives, but daily habits—especially consuming sugary drinks and snacks—often go unnoticed and untracked. Our surveys showed that most respondents consume at least one sugary item every day, rarely check nutrition labels, and don’t monitor how these habits affect their health.

At the same time, existing health apps either make it too hard to log food, focus on long-term goals like weight loss, or ask for too much data too soon. This leaves a big gap: young users lack a simple way to see how their sugar choices affect their body *right now* and to get quick suggestions that help them make better choices the next time.

Why This Matters

Frequent sugar intake is linked to long-term risks such as poor sleep, low energy, weight gain, and conditions like prediabetes and diabetes. When users don’t get immediate feedback or easy, actionable tips, they don’t build healthy habits, and the risk silently grows.

If we can solve the real-time feedback loop—capture a sugar choice quickly, meaningfully explain its impact, and suggest a tiny next step—we can help people reduce sugar habits before serious health issues develop.

1.2 Primary Goal

Build a prototype solution that helps users reduce sugar-related health risk behaviors by providing real-time, personalized, context-aware recommendations based on their daily inputs and passive data.

Core Requirement: Participants should not only build a functional prototype; they must design a *habit loop* that makes users want to come back every day to log their sugar choices.

Chapter 2

System Objectives & Features

2.1 Fast, Frictionless Sugar Event Capture [Mandatory]

Objective: Build a web/app interface that allows users to log a sugary choice in under 10 seconds without breaking flow.

What participants must build

Users must be able to log sugar using any one of the following:

- **One-tap presets:** (e.g., chai, sweets, cold drink, packaged snack).
- **Photo-based logging:** (Optional).
- **Voice-based logging:** (Optional).

Expected Outcome

- Users can log sugar events quickly and naturally.
- No drop-off due to friction or complexity.

2.2 Signup-Free, Gamified Onboarding [Mandatory]

Objective: Allow users to start without an email or mobile number using a device-based anonymous user ID.

Data Collection

Collect only the minimum required data:

- Age (via DOB picker or age picker).
- Height.
- Weight.
- Gender.
- **Note:** BMI must be calculated silently, never explicitly asked.

How Onboarding Should Feel

It should feel like a game, not a form:

- One question per screen.
- Use sliders, cards, and simple animations.
- Show a progress indicator (e.g., “3 of 5 completed”).

2.3 Passive Health Data Sync [Mandatory]

Objective: Ask permission in a friendly way to access health data.

- **Data points:** Step count, Heart rate (if available), Sleep duration (optional).
- **Rule:** Data should be used only to improve insights. Never show raw biometric numbers to users.
- **Purpose:** This data is used only to make responses more context-aware and precise.

2.4 Daily Ritual & Habit Formation [Mandatory]

Objective: Create a daily ritual, similar to Duolingo’s “Daily Lesson”.

- User should complete a sugar log in < 10 seconds.
- User should feel motivated to return every day.

Gamification Rules

- **Daily Streaks:** Day 1, Day 3, Day 7, Day 30.
- Visual progress bar.
- Friendly reminder copy (e.g., “Log today to protect your streak”).

2.5 Immediate Feedback & Rewards [Mandatory]

After every sugar log or action, the system must:

1. Show a visual animation.
2. Play a short success sound.
3. Give one reward (XP/Points, Badge, or Small surprise reward).

Variable Rewards

Not every action gives the same reward. Some days feel more rewarding than others. This creates curiosity and repeat engagement (similar to Blinkit / Swiggy offers).

2.6 Optional Signup for Enhanced Experience

Rule: Clear separation must exist between the Free (anonymous) experience and the Logged-in (enhanced) experience.

- **When to ask:** Only after value is delivered. Never force signup at the start.
- **What it unlocks:** Extra rewards, deeper personalization, history across devices, advanced features.
- **How it should feel:** Not forced. Framed as an upgrade: “Want deeper insights & rewards? Unlock now.”

2.7 Context-Aware Insight Generation (ML-Based)

Objective: Use collected data (age, BMI, steps, sleep, time of day, etc.) to generate an immediate, personalized insight explaining the short-term impact of today’s sugar choice.

Insight Characteristics

- Simple language.
- Cause → Effect format.
- Non-medical, non-diagnostic.
- *Example:* “On days like this, sugar often reduces your sleep quality.”

2.8 Personalized Corrective Action Suggestion

Based on context, suggest **one** immediate action:

- 10-minute walk.
- Drink water.
- Protein snack swap.

Rules

- One primary action only.
- Action must be doable immediately.
- Action choice should adapt based on Age, Activity level, Steps, and Sleep.

2.9 Gamified Scoring System **[Mandatory]**

Points awarded dynamically creates optimization behavior and encourages better timing and consistency.

Examples

- Successful walk: **+5 points**
- Logging before 6pm: **+3 points**
- Completing action within 30 minutes: **+7 points**

Chapter 3

Psychological Framework & User Experience

3.1 Psychological Principles Used

The solution must utilize the following principles to drive user behavior.

Principle	How it is Applied
Loss Aversion	Streaks that users don't want to lose.
Instant Gratification	Immediate animation + sound after logging.
Variable Rewards	Surprise bonuses (users don't know what they will get).
Commitment	Optional signup asked only <i>after</i> value is provided.
Habit Formation	Creating a daily ritual loop.

3.2 Final Expected Experience

A successful solution should make users feel:

“This is fast, fun, and something I want to do every day.”

3.3 Bonus Objectives (Extra Points)

Participants can earn bonus recognition by:

- Implementing ML-driven decision personalization.
- Adding voice or photo logging.
- Using high-quality animations and sounds.
- Showing explainability of why a suggestion was chosen.

Chapter 4

Submission & Judging

4.1 Submission Rules & Guidelines

To ensure a level playing field and high-quality output, the following rules are mandatory:

1. **Continuous Development Rule:** After the first 5 hours of the hackathon, teams must commit code to their repository at least once every 60 minutes. This is to track progress and ensure transparency.
2. **Tech Stack:** Preference will be given to projects using latest technologies (e.g., modern cross-platform frameworks, edge ML, or advanced UI/UX libraries).
3. **The “Value-First” Signup:** You may only prompt for a full user signup after the user has experienced the core value of the app.

4.2 Judging & Scoring Criteria

Gamification & Retention	Use of streaks, variable rewards (XP/badges), animations, and psychological “hooks”.
Feature Implementation	Smoothness of logging, passive data sync, and the “Signup-Free” flow.
ML Integration	Quality and relevance of the context-aware insights and corrective suggestions.
Innovation & Creativity	Originality of the UI, “feel-good” micro-interactions, and creative problem-solving.

Hints

Think about why Duolingo is addictive or why you often spend more on Blinkit/Swiggy. Apply those same principles here.